Assignment: Super Resolution

Task 1: Build, Train, and Test an SRCNN

- Implement the SRCNN architecture.
- Train the network using a suitable super-resolution dataset.
- Test the trained model on unseen low-resolution images.
- Record the training and testing performance.

Include:

- SRCNN architecture diagram
- Training and validation loss curves
- Test results
- Model summary (number of parameters)

Task 2: Build, Train, and Test an ESPCN

- Implement the ESPCN architecture.
- Train the model using the same dataset and training settings used for SRCNN.
- Test the trained model on unseen images.

Include:

- ESPCN architecture diagram
- Training and validation loss curves
- Test results
- Model summary (number of parameters)

Task 3: Build, Train, and Test an SRResNet

- Implement the SRResNet architecture.
- Train the network using a suitable super-resolution dataset.
- Test the trained model on unseen low-resolution images.
- Record the training and testing performance.

Include:

- SRResNet architecture diagram
- Training and validation loss curves
- Test results
- Model summary (number of parameters)

Task 4: Build, Train, and Test an SRGAN

- Implement the SRGAN architecture.
- Train the network using a suitable super-resolution dataset.
- Test the trained model on unseen low-resolution images.
- Record the training and testing performance.

Include:

- SRGAN architecture diagram
- Training and validation loss curves
- Test results
- Model summary (number of parameters)

Task 5: Build, Train, and Test an EDSR

- Implement the EDSR architecture.
- Train the model using the same dataset and training settings used for SRGAN
- Test the trained model on unseen images.

Include:

- EDSR architecture diagram
- Training and validation loss curves
- Test results
- Model summary (number of parameters)

Task 6: Compare Five NNs Using Smartphone Images

1. Capture at least 10 high-resolution images using your smartphone.
2. Convert each image into a low-resolution version using bicubic downsampling.
3. Apply SRCNN, ESPCN, SRResNet, SRGAN and EDSR to reconstruct high-resolution images.
4. Compare the reconstructed images with the original high-resolution images.

For each image, report:

- Original HR image
- Low-resolution image
- SRCNN output
- ESPCN output
- SRResNet output
- SRGAN output
- EDSR output

Evaluate the models using:

- PSNR (Peak Signal-to-Noise Ratio)
- SSIM (Structural Similarity Index)
- Inference time
- Number of parameters
- Visual quality

Task 6: Analysis and Discussion

Compare five super resolution models by discussing:

- Reconstruction quality
- Preservation of edges and textures
- Computational efficiency
- Model complexity
- Training time
- Testing speed
- Memory usage

Based on both quantitative metrics and visual inspection, determine:

Which neural network produces better super-resolution images? Explain your conclusion with appropriate evidence.